

Huzaifa Ahmad

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EDUCATION

University of Victoria

Victoria, BC

Bachelor of Engineering in Computer Engineering

Expected Apr 2028

- **Relevant coursework:** Digital Design, Microprocessor Systems, VLSI Design, Signals & Systems, Electronic Circuit Analysis & Design, Analog Electronics

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, VHDL, MATLAB, Assembly

Frameworks & Tools: React, Git, Linux, FreeRTOS, MicroPython, Xilinx Vivado

Embedded & Digital: STM32 (Cortex-M), RP2040, Basys-3 FPGA, SPI, I2C, UART, ADC, PWM, bare-metal firmware

Hardware & Lab: KiCad, Altium Designer, schematic capture, PCB layout, surface-mount (SMD) soldering, board bring-up, oscilloscopes, multimeters

EXPERIENCE

UVic Satellite Design Club

Aug 2024 – Present

Hardware Engineer, On-Board Computer

Victoria, BC

- Contributed on-board computer hardware to MARMOTSat, a **3U** ionospheric research CubeSat built by nearly **60** UVic contributors and launched on SpaceX's Transporter-17 in July 2026.
- Selected **40+** components by checking electrical, power-dissipation, thermal, and footprint requirements against manufacturer datasheets, finalizing the on-board computer's bill of materials (BOM) for production.
- Hand-assembled prototype boards by reflow soldering **0805**, **0603**, and SOIC parts, then verified power rails and signal continuity on the bench before handing them off for subsystem integration.
- Wrote a repeatable assembly and bring-up test procedure that other members used to test later board revisions.

UVic Rocketry

Jun 2023 – Present

Software Developer, Ground-Support Telemetry

Victoria, BC

- Contributed ground-support telemetry software to Andúril, UVic's Launch Canada rocket series, including Andúril-2, which placed **5th** at Launch Canada 2025 after reaching **Mach 2.2** and **32,694 ft**.
- Developed the sensor-configuration UI for the team's ground-support telemetry app in React and TypeScript, used by the avionics team to set parameters for **6** onboard modules, including the inertial measurement unit (IMU), barometer, airbrakes, and strain gauges, during pre-launch testing.

PROJECTS

Custom PCB Design | Altium Designer

May – Jun 2026

- Produced a design-rule-check (DRC) clean manufacturing package with Gerbers, ODB++, BOM, and assembly drawings by taking a board through the full Altium Designer flow of schematic capture, placement, routing, and copper pours.

Pipelined CPU | VHDL, Xilinx Vivado, Basys-3 FPGA, Assembly

Jan – Apr 2026

- Designed a **16-bit**, **5-stage** pipelined CPU in VHDL with a custom instruction set, ALU, control unit, and hazard detection, validated with register-transfer level (RTL) testbenches.
- Met all static timing constraints at **50.8 MHz** on a Basys-3 FPGA while using under **5%** of the device's resources.

Real-Time OS Traffic Light Controller | C, FreeRTOS, STM32F4, GPIO, ADC

Jan – Mar 2026

- Programmed a multi-task FreeRTOS application on an STM32F4 that drives **19** LEDs through **3** daisy-chained shift registers, using message queues and timers between tasks and ADC input to set cycle timing.
- Gave the light-control task top priority and passed all inter-task data through queues.

FM Radio-Clock PCB | KiCad, RP2040, MicroPython, SPI, I2C

May – Aug 2024

- Prototyped and brought up a hand-assembled **2-layer** PCB with an RP2040, RDA5807 FM tuner, LM386 audio amplifier, SSD1306 OLED, and rotary-encoder controls on regulated **3.3 V** and **5 V** rails.
- Debugged the board with an oscilloscope and multimeter, then implemented MicroPython drivers over SPI and I2C for the clock display, alarms, FM tuning, and volume control.